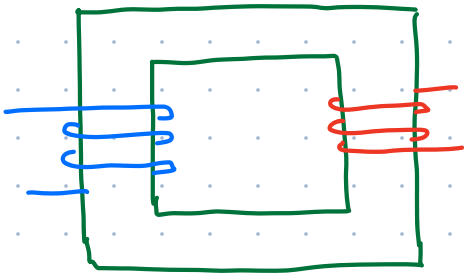
Review 4 (HW 4) 09/28/2026

Coupled coils

Dot convention: „ currents entering dots throw flux in the same direction".

We want our reference model to work irrespective of winding direction

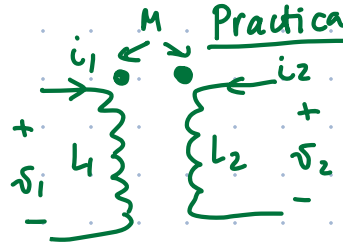


Ideal



$$\frac{i_1}{-i_2} = \frac{N_2}{N_1} \qquad \frac{v_1}{v_2} = \frac{N_1}{N_2}$$

Practical



$$\lambda_1 = N_1\phi_1 = L_1 i_1 + M i_2$$

$$\lambda_2 = N_2\phi_2 = M i_1 + L_2 i_2$$

$$v_1 = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

$$v_2 = M \frac{di_1}{dt} + L_2 \frac{di_2}{dt}$$

⇓

phasors:

$$\bar{V}_1 = j\omega \bar{I}_1 L_1 + j\omega \bar{I}_2 M$$

$$\bar{V}_2 = j\omega \bar{I}_1 M + j\omega \bar{I}_2 L_2$$

① Fix ref. frame
② Apply ref. model
③ Subst. back

Coefficient of coupling: $0 \le K \le 1$

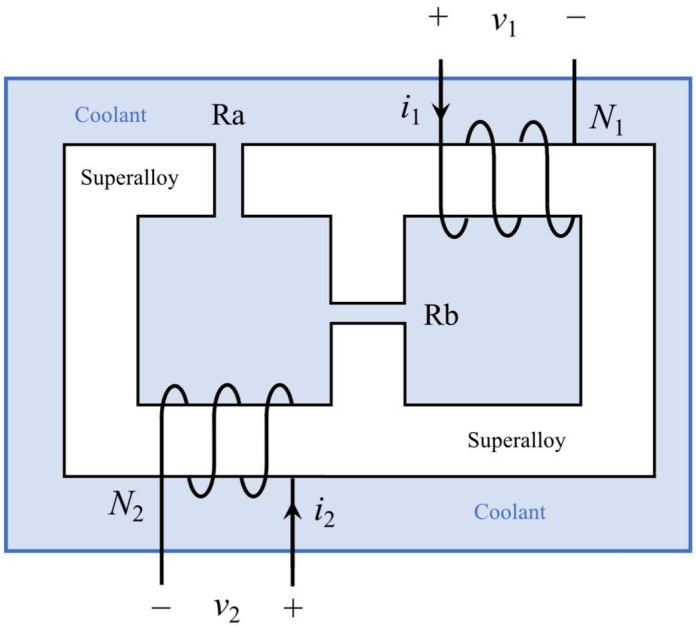
$$K = \frac{M}{\sqrt{L_1 L_2}}$$

① Spring 2021, M1, pr 4

Problem 4 (25 Points)

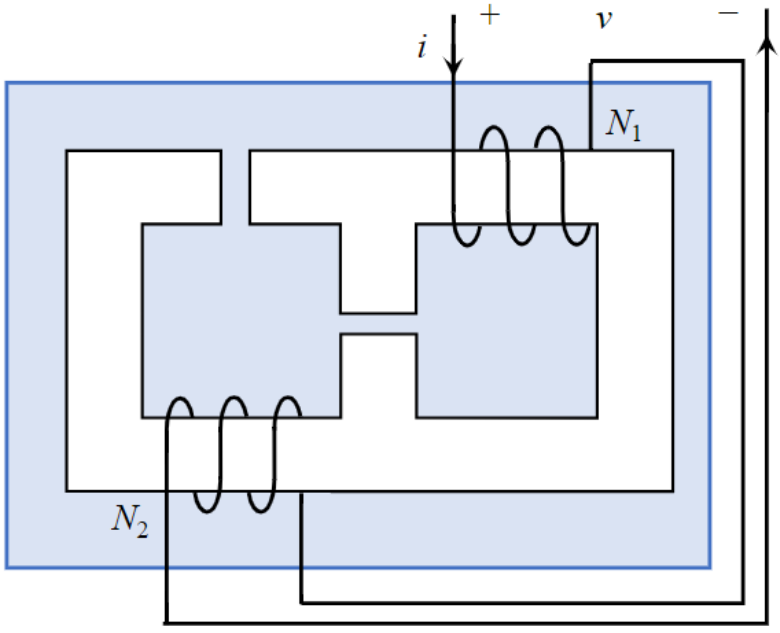
You have been hired by a stealth-mode startup to develop a superconducting transformer for a possible mission to Mars.

The transformer is built out of magnetic super alloy, and immersed in a liquid coolant. The super alloy is so magnetically permeable that its permeability is essentially infinite ($\mu_r \rightarrow \infty$) while the coolant is a factor of five times more permeable than vacuum ($\mu_r = 5$). The two coils are connected to a resonant power converter circuit with the voltage and current labels as shown. The first coil has $N_1 = 20$ turns while the second coil has $N_2 = 10$ turns.



a) We computed Ra = 200 At/Wb and Rb = 800 At/Wb by neglecting the effects of fringing. Use these figures to compute the mutual inductance between the two coils, and state the unit.

(c)



In part c, two coils are connected in series as shown. What is the combined inductance of the structure?

$L_1 = 2.5\,H$

$L_2 = 0.5\,H$

$M = 1\,H$